

Going Beyond Usability and UX: Adding Dependability, Safety and Security to Interactive Systems and Interactive Technologies

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ABSTRACT

This course takes both a practical and theoretical approach to introduce the principles, methods and tools for including “non-standards” properties inside interactive systems and interaction technologies. The course focusses on safety, dependability and security but encompasses additional contributing factors such as privacy, availability and trustworthiness and relate them to HCI-centric properties such as User Experience and Usability. The course will cover design, implementation and evaluation activities to assess these properties. A special attention will be made on how the properties are competing with one another in terms of knowledge of designers and resources required at development time. To paraphrase Susan Dray: “if it is not usable, it does not work” this course will argue that “if it does not work (i.e. it is not dependable) it is not usable” as well as “if it is not safe, it is not usable”. The concepts in this course will be supported by multiple concrete examples taken from industries including autonomous cars, aircraft cockpits, large command and control systems but also research labs prototypes targeting at deploying technologies in real-life settings.

CCS CONCEPTS

• **CCS CONCEPTS**; • **Human-centered computing**; • **Hardware** → Robustness; Fault tolerance; Robustness; Safety critical systems; • **Software and its engineering** → Software organization and properties; Extra-functional properties; Software usability; Software organization and properties; Extra-functional properties; Software safety;

KEYWORDS

Additional Keywords and Phrases: Usability, User Experience, Safety, Dependability, Interactive Systems

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1 INTRODUCTION

With the early work on understanding interactive systems [1] came the identification of properties that “good” interactive systems should exhibit (e.g. honesty) and “bad” properties that they should avoid (e.g. deadlocks). Later, guidelines for the design of interactive systems [2] were provided, identifying in a similar way “good” property (e.g. guidance), in order to favor usability of these systems. In the area of software engineering, early work [3] identified two main good properties of computing systems namely safety (i.e. nothing bad will ever happen) and liveness (i.e. something good will eventually happen). In [4] a hierarchy of software properties is proposed identifying, for the first time, explicit relationships between properties gathered in a hierarchy (e.g. “reactivity” divided in “recurrence” and “persistence”). While in the area of Human-Computer Interaction the properties were initially expressed in an informal way, [5] and [6] proposed the use of temporal logics to describe these properties.

In HCI the focus was early made on usability standardized in the ISO 9241-11 [7] which decomposed usability in effectiveness, efficiency and satisfaction. More recently, User Experience [8] was introduced which includes affects and emotions, aesthetics or values. [10] further decomposes this property into six contributing factors: Aesthetics, Emotions, Identification, Stimulation, Meaning and Value and Social-connectedness.

As a consequence, most of the contributions in the area of HCI focus on these two properties, leaving aside other (more system-related properties) properties such as dependability [13], safety [13] or security [12] and privacy [11]. As an example, dependability is decomposed into five contributing factors: availability, safety, reliability, integrity and maintainability while security is decomposed into three: availability, integrity and privacy. One key element that can be seen from these decompositions of properties is that they are intertwined. Indeed, two of the three factors of security are shared with dependability. Work such as [14] and [15] has been dedicated to making explicit the articulation between these properties, including the HCI-related ones: usability and user experience.

The key aspect is that some HCI research touches systems for which dependability or safety (just to name two) are of prime importance. Work in [16] introduces a new interaction technique for interactive cockpits of aircrafts but the dependability is critical and not address in that paper but fortunately presented in another venue dedicated to safety and dependability [17]. On a similar viewpoint, as usability-driven research proposes new interactive technologies such as multitouch ones [23] research has to be conducted on the dependability side to propose new formal approaches to engineer those interactive technologies [9]. As for safety, [18] introduces an AR application to support decision making while cycling. In that

paper safety is mentioned during the evaluation but no relationship to safety engineering practice is mentioned.

This course builds on top of this perspective adopting a multi-properties perspective and considering interactive systems as artefacts supporting operators' work and users' tasks and for which each property has to be assessed in the light of the other conflicting ones. We will consider that the audience is knowledgeable in usability and user experience, thus the focus will be on safety, dependability and security and their interplay with usability and UX. We believe that this will increase the potential deployment in real settings of HCI research that remains otherwise disconnected from some of the real-world constraints.

2 CONTRIBUTION AND BENEFIT

This course will provide newcomers with background knowledge in automation. It provides an overview on how the recent advances in safety, dependability and security engineering can be exploited to design and assess interactive applications and interaction techniques. This course will also provide material for lecturers who want to include properties assessment in their HCI courses. Finally, the course will provide examples from various domains including Aviation, Air Traffic Management, Automotive, Nuclear Power Plants . . . highlighting both successes and failures of properties assessment. Concrete examples and exercises will also be provided to support the application of the course elements to practical cases.

3 OBJECTIVES

On completion of this tutorial, attendees will:

- Know the intrinsic and extrinsic characteristics of the properties of interactive systems and interaction technologies,
- Be able to decompose these properties in contributing factors and to assess the presence of these contributing factors in a prototype or system,
- Know many examples on how multiple properties have been used in various domains such as automotive, aviation, Air Traffic Control, space domain . . . ,
- Know and understand the cost (in terms of time and resources) of considering a given property,
- Understand the potential consequences for not considering relevant properties in system design and engineering,
- Know the various disciplines and their research venues for each of the presented properties.

4 AGENDA

This course is composed of one single course unit going from conceptual to very practical. It includes interactive hands-on exercises, case studies and practice on properties identification and assessment in the content of interactive systems design.

5 AUDIENCE AND PREREQUISITE

This course is open to researchers, practitioners, educators and students of all experience levels. No specific skills or knowledge are required beyond a background in User Centered Design. Paper [13] should be read before attending the lecture.

6 COURSE BACKGROUND

Four related courses have been given at CHI, but content and philosophy of this one is very different. This "fifth" edition considers the feedback from participants of the previous editions and thus focusses on the properties only briefly mentioned in the Automation-related courses. The first course was about "Designing and Assessing Interactive Systems Using Task Models" (given at CHI 2015 [19] was using automation as a design alternative for tasks allocation). Feedback from participants showed that they would have a strong interest in automation per se (both foundations and applications) with a focus on autonomous cars. A second course was proposed at CHI 2018 [20] about the risks and benefits of higher levels of automation and was entitled "Automation: Danger or Opportunity? Designing and Assessing Automation for Interactive Systems". A Third course (overnight at CHI 2021 [21]) entitled "Introductory Course on Automation and its Use for Interactive Systems Design and Engineering" focused on engineering aspects and the foundations of automation. The fourth course (in person at CHI 2022 [22]) entitled "How to Design and Assess Automation for Interactive Applications and Interaction Techniques" focused on automation in interactive technologies (usually hidden and not designed with the required attention and knowledge). Again, feedback from attendees showed another interest in the foundations of safety, dependability and security as basic concepts for first time encounters.

This fifth course focuses on the use of these properties and their relationship with UX and Usability, which was only touched by some examples of successes (for instance transfer functions in input devices) and failures (for instance autonomous cars). A longer version of this course (with a strong focus on automation) it taught at Eurocontrol.

7 PRACTICAL ACTIVITIES

Various exercises will be given to the attendees. Exercises will consist in analyzing some interactive technologies published at HCI conferences e.g. technology for command and control or cycling. Then the attendees will be faced with design activities proposing extensions to ensure safety, security or dependability.

Two case studies from the real world (interactive cockpits and rocket launch pads) will be studied in depth showing how conflicting properties are handled in real contexts.

8 INSTRUCTOR'S BACKGROUND

The course instructor is professor in Computer Science at the University Toulouse 3. Since the early 90's his research focusses on interactive systems engineering proposing notations, methods and tools to integrate multiple properties including usability, UX, dependability and safety. He is working on safety critical systems and their development processes with various industries (e.g. Airbus, Eurocontrol). He is a member of CHI, INTERACT and EICS steering committees and a member of CHI academy. He is also a member of the program committees of conferences from the domains of Safety, Dependability and formal methods (e.g. SAFECOMP 2023, DSN 2014 and ABZ 2023).

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